Training Configuration and Summary:

Dataset used: rssi_odom_dataset_11apr25.parquet

LSTM Multi-Step Training Report

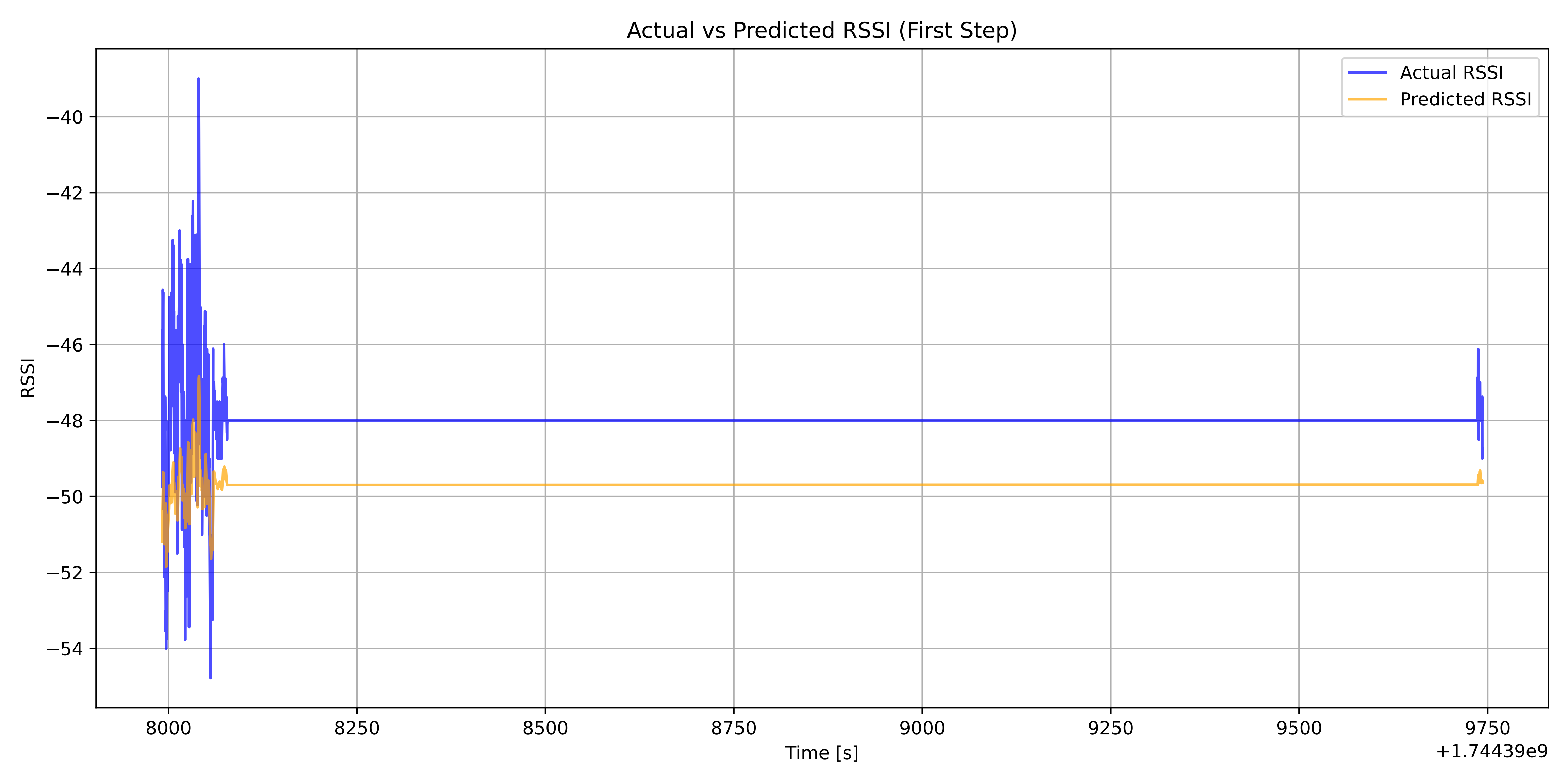
Report Generated: 2025-05-07 08:25:53

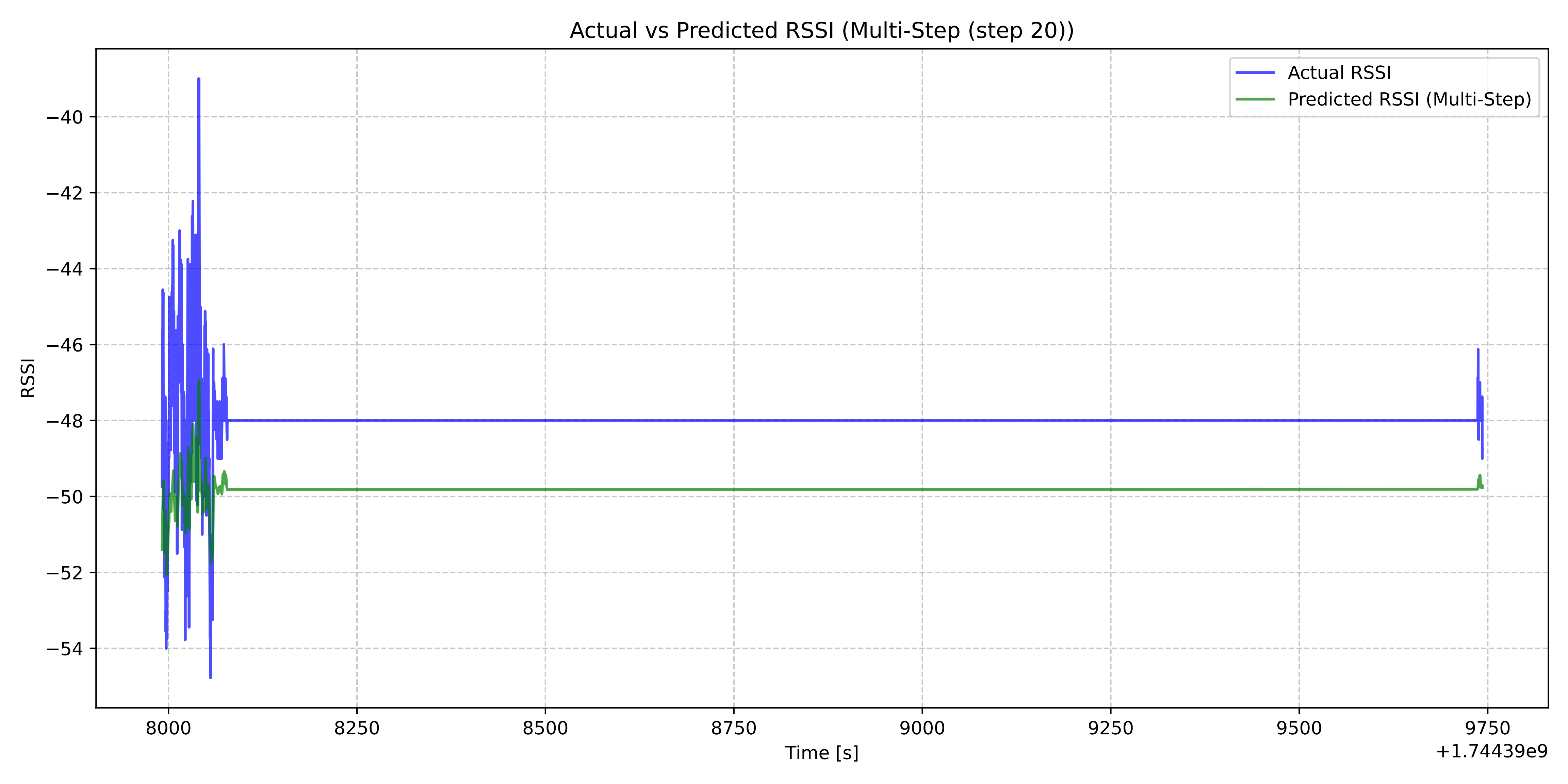
Training Start: 2025-05-06_19-53-30
Training End: 2025-05-06_19-57-03
Training Duration: 3.54 minutes

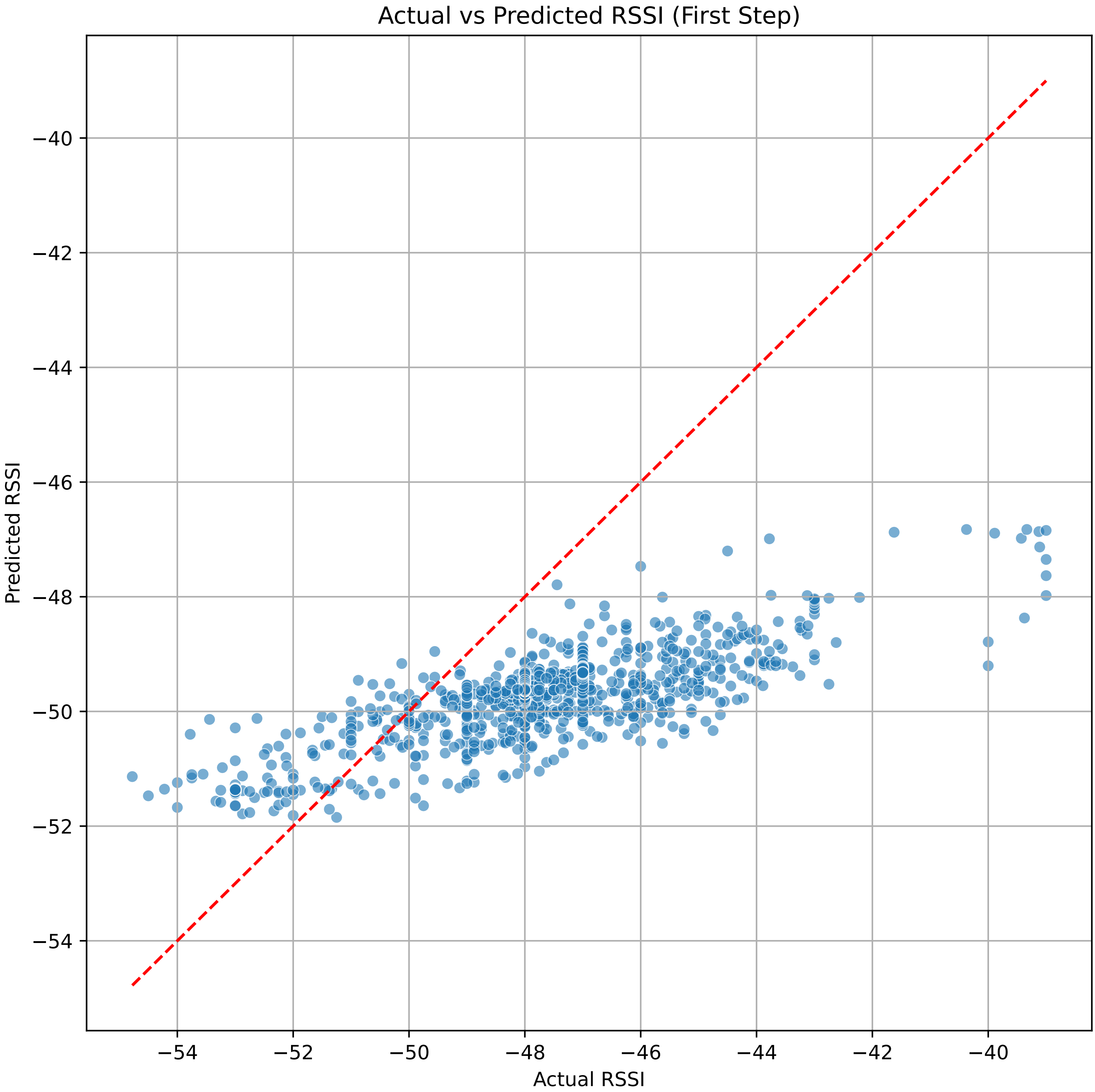
Best Hyperparameters:
Number of Layers: 2
Activation: relu
Optimizer: Nadam
Dropout Rate: 0.20

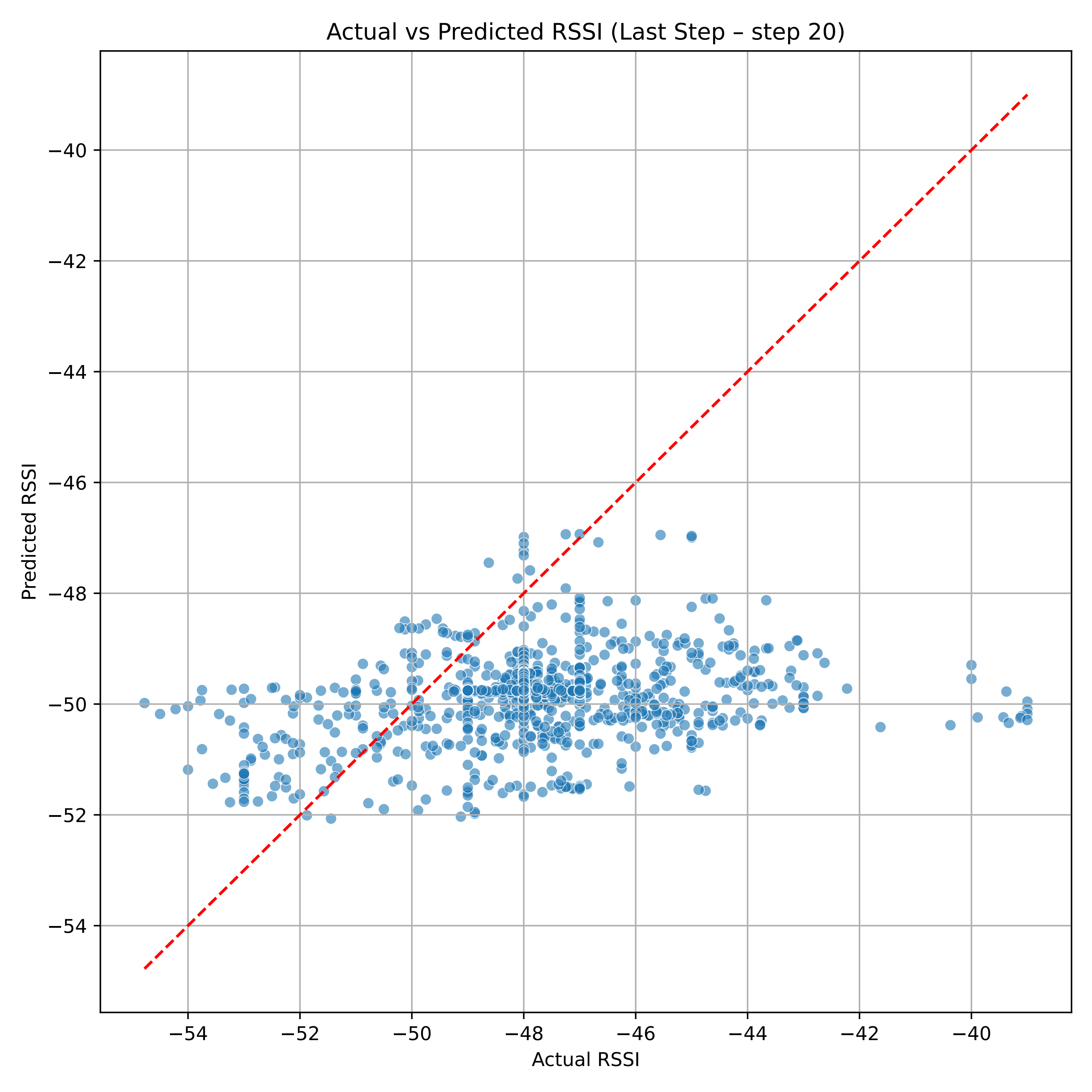
MAE: 2.4606
MSE: 9.0943
RMSE: 3.0157
R2 Score: -0.5900

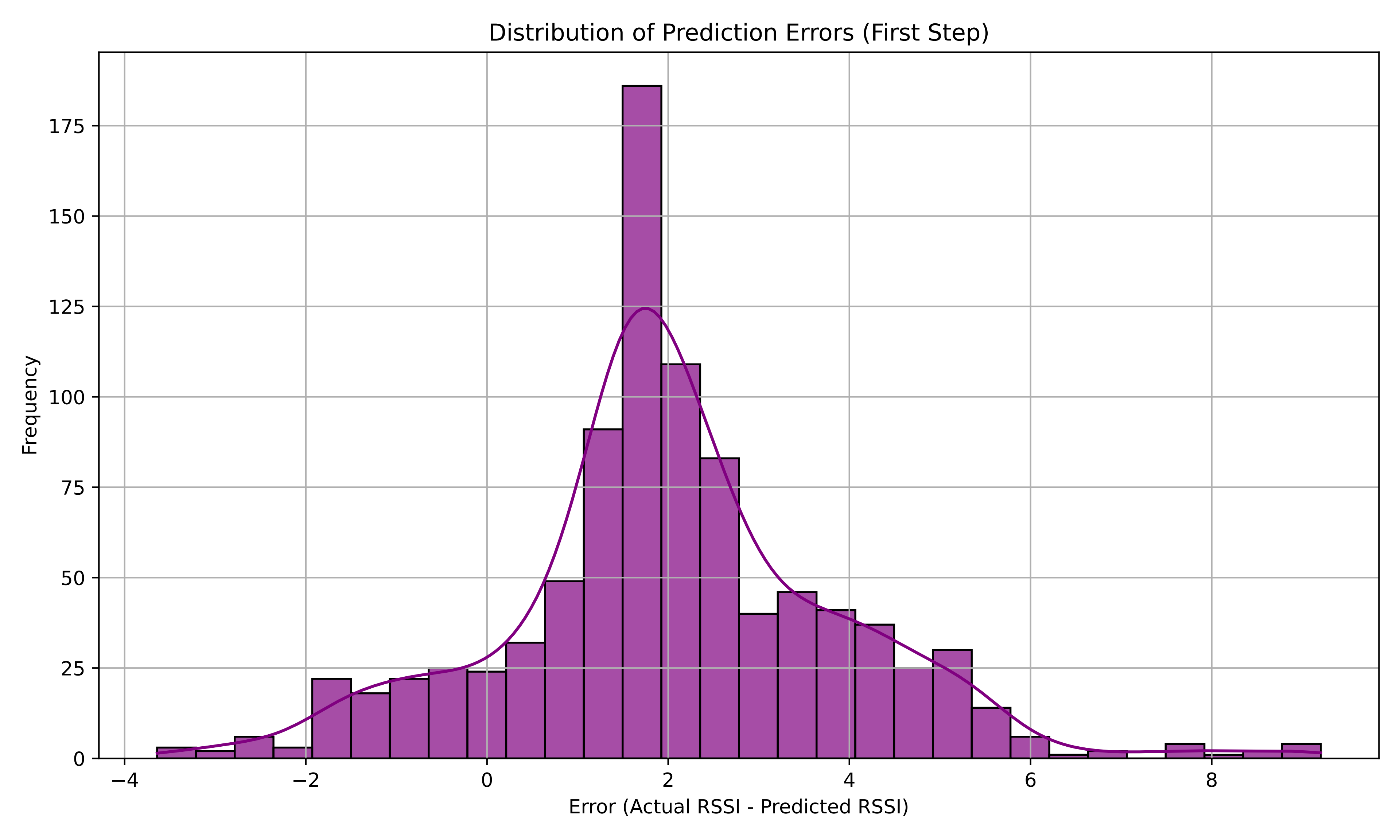
Multi-Step Forecast:
Forecast Steps: 20
MAE: 0.0867
MSE: 0.0106
RMSE: 0.1031
R2 Score: -0.3207


Actual vs Predicted RSSI (First Step)
Actual RSSI
Predicted RSSI
RSSI
−40
−42
−44
−46
−48
−50
−52
−54
8000
8250
8500
8750
9000
9250
9500
9750
+1.74439e9
Time [s]

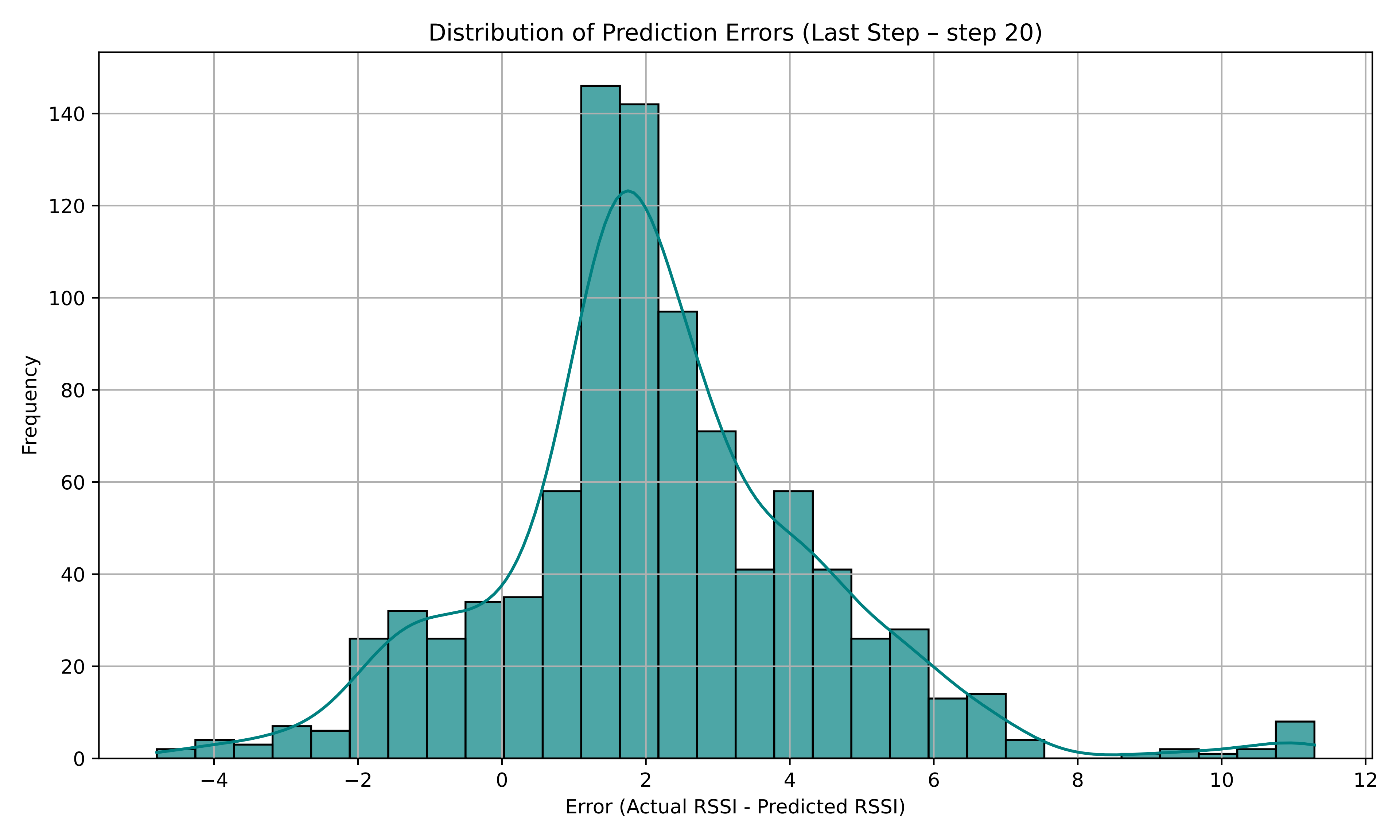

Actual vs Predicted RSSI (Multi-Step (step 20))
Actual RSSI
Predicted RSSI (Multi-Step)
RSSI
−40
−42
−44
−46
−48
−50
−52
−54
8000
8250
8500
8750
9000
9250
9500
9750
Time [s]
+1.74439e9

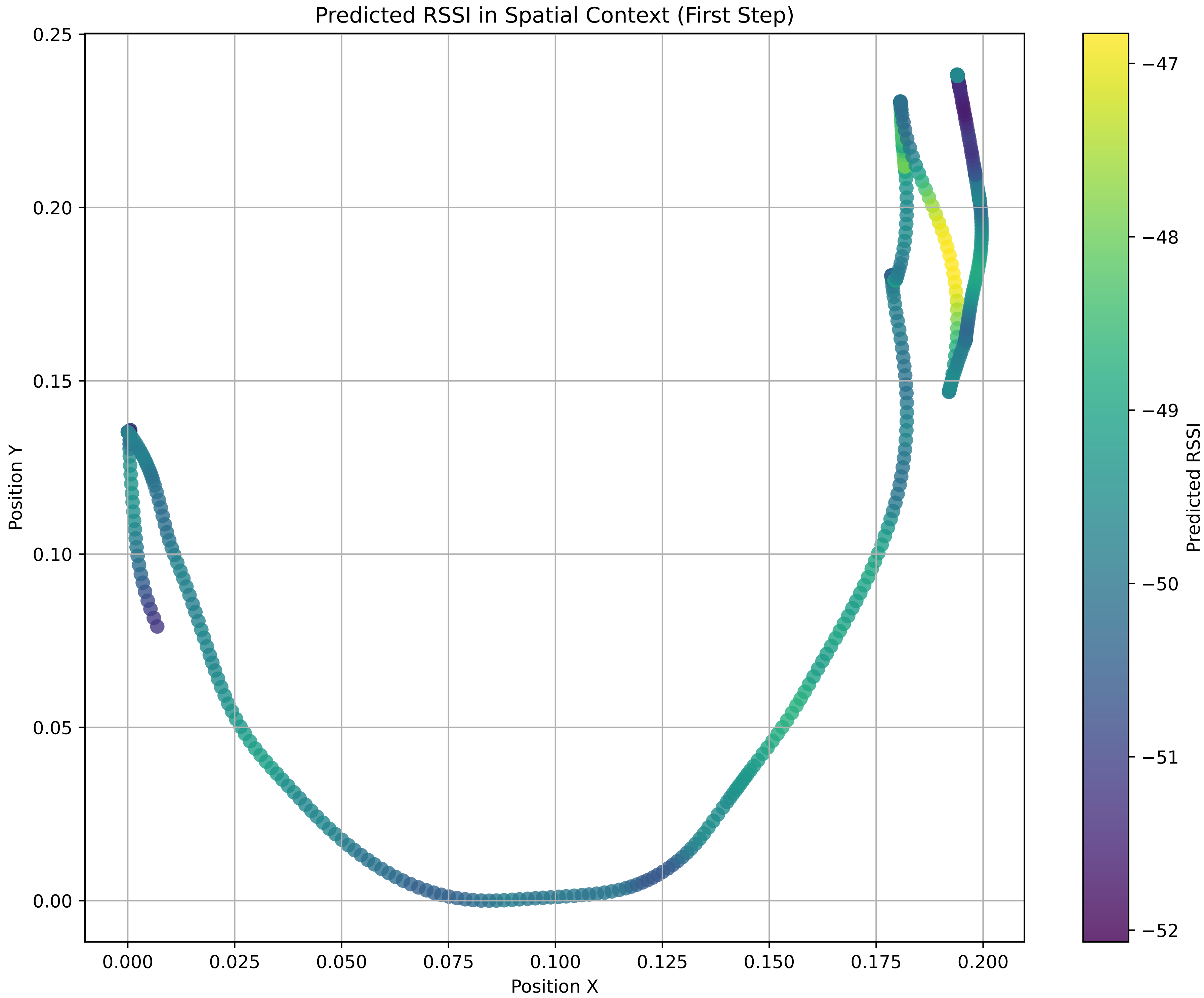

Actual vs Predicted RSSI (First Step)
Predicted RSSI
Actual RSSI
−40
−42
−44
−46
−48
−50
−52
−54
−54
−52
−50
−48
−46
−44
−42
−40

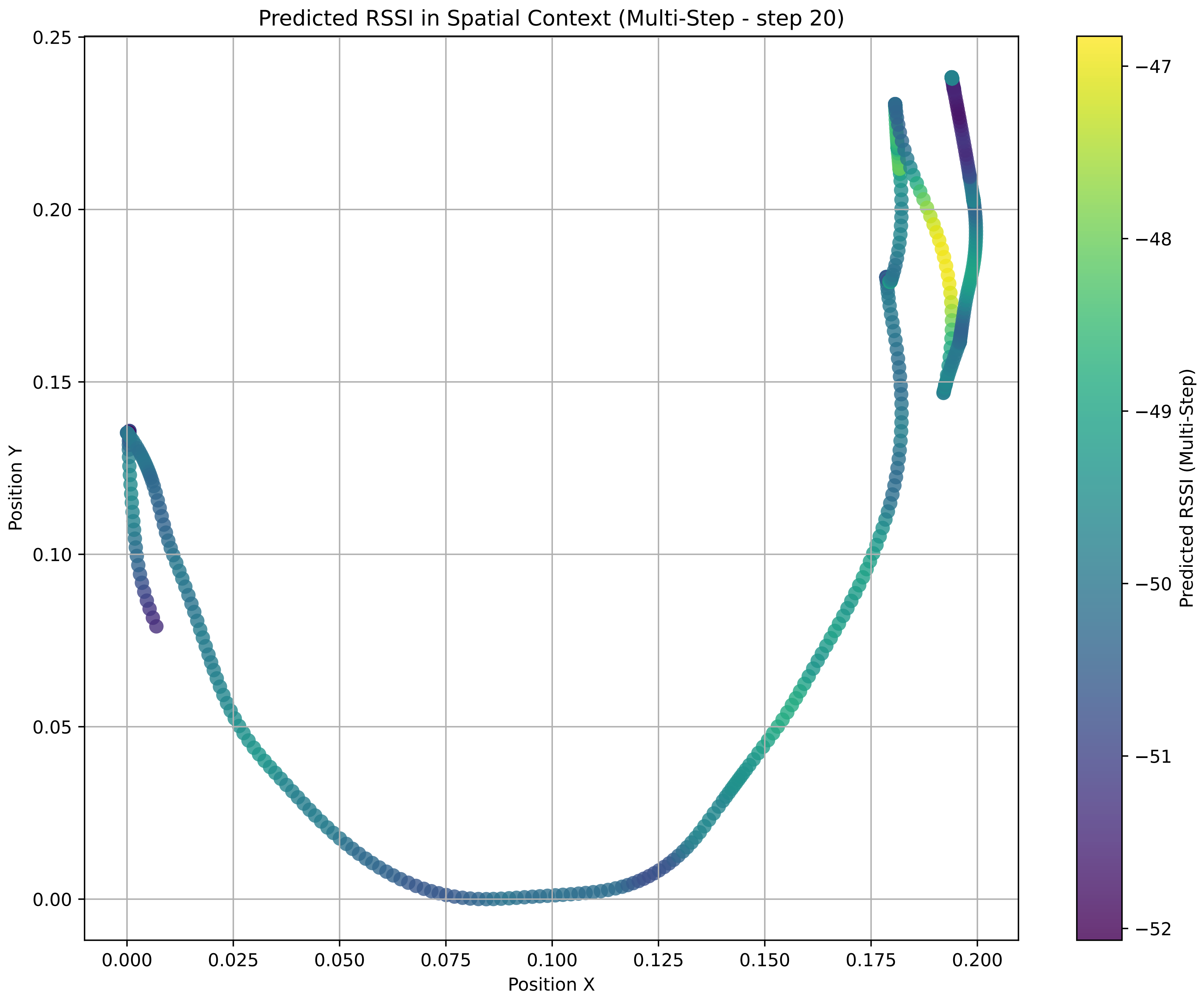

Actual vs Predicted RSSI (Last Step – step 20)
Predicted RSSI
Actual RSSI
−40
−42
−44
−46
−48
−50
−52
−54
−54
−52
−50
−48
−46
−44
−42
−40



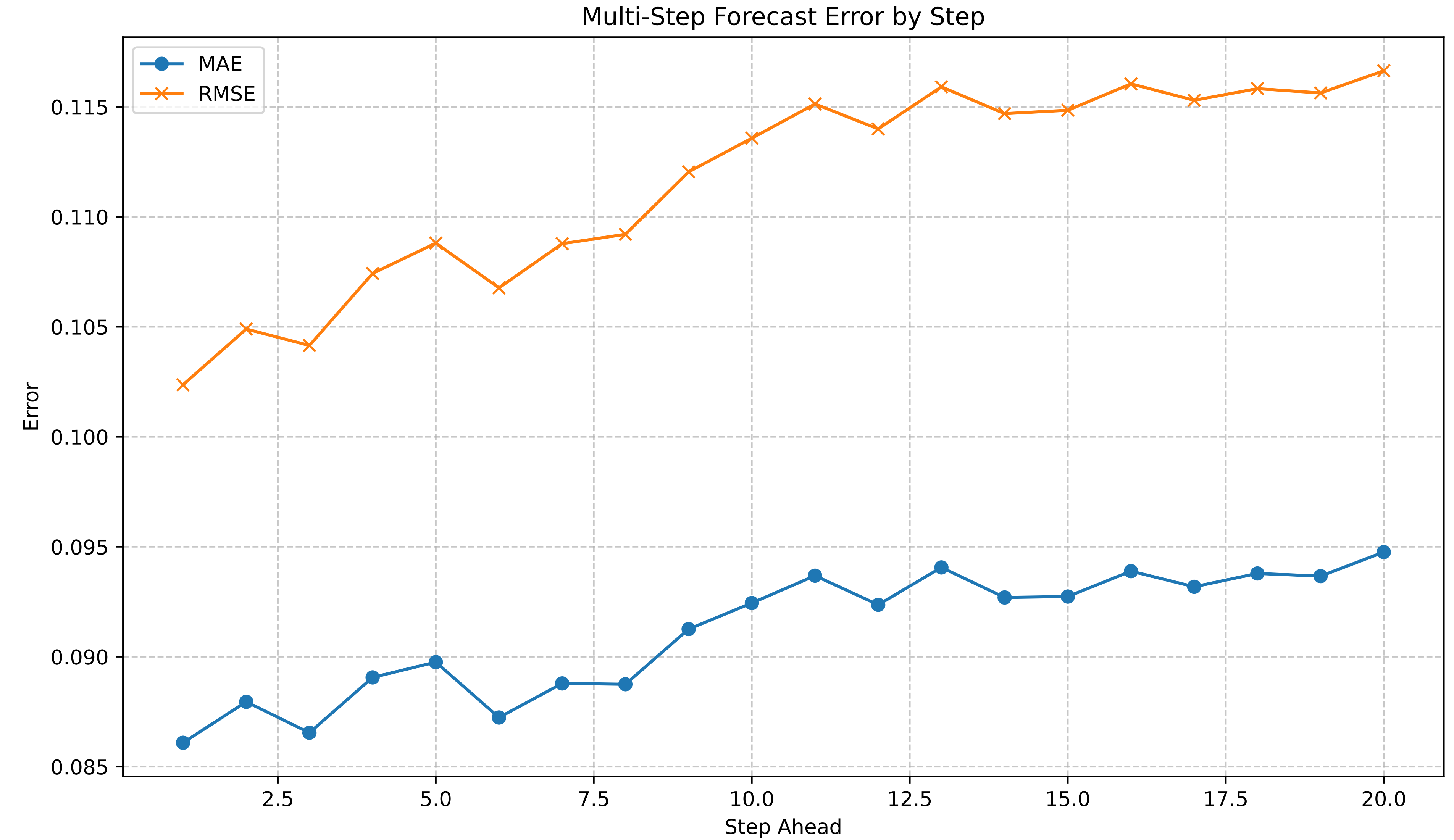
Distribution of Prediction Errors (First Step)
Frequency
Error (Actual RSSI - Predicted RSSI)
0
25
50
75
100
125
150
175
−4
−2
0
2
4
6
8


Distribution of Prediction Errors (Last Step – step 20)
Frequency
Error (Actual RSSI - Predicted RSSI)
0
20
40
60
80
100
120
140
−4
−2
0
2
4
6
8
10
12


Predicted RSSI in Spatial Context (First Step)
Position Y
Position X
0.25
0.20
0.15
0.10
0.05
0.00
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Predicted RSSI
−47
−48
−49
−50
−51
−52



Predicted RSSI in Spatial Context (Multi-Step - step 20)
Position Y
Position X
0.25
0.20
0.15
0.10
0.05
0.00
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Predicted RSSI (Multi-Step)
−47
−48
−49
−50
−51
−52



Multi-Step Forecast Error by Step
MAE
RMSE
Error
Step Ahead
0.085
0.090
0.095
0.100
0.105
0.110
0.115
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

```
Resumo Automático
-----------------
Dataset:        rssi_odom_dataset_11apr25.parquet
Modelo:         LSTM
Forecast steps: 20

MAE / RMSE (dB)
    • 1º passo : 0.09 / 0.10
    • Mediana  : 0.09 / 0.11
    • 20º passo : 0.09 / 0.12

Crescimento do erro: 10.1%   →   Estável (erro cresce < 20 %)

3 piores passos (MAE): 20 (0.09), 13 (0.09), 16 (0.09)

Recomendação:
O modelo é adequado para decisões de hand-over até o horizonte máximo
configurado.
```